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Improving Sim-to-Real transfer of Learning-based Policies for Quadcopter Racing through Domain Randomization

Semestral Project

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Acknowledgments

Firstly, I would like to express my gratitude to my supervisor.

Swap this for the Thesis Assignment, when you have it from the department.

Declaration

I declare that presented work was developed independently, and that I have listed all sources of information used within, in accordance with the Methodical instructions for observing ethical principles in preparation of university theses.

Date
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Abstract

The study of autonomous Unmanned Aerial Vehicles (UAVs) has become a prominent sub-field of mobile robotics.

Keywords Unmanned Aerial Vehicles, Automatic Control

Abstrakt

Výzkum na poli autonomních bezpilotních prostředků (UAV) se stal významným oborem mobilní robotiky.

Klíčová slova Bepilotní Prostředky, Automatické Řízení

Abbreviations

UAV Unmanned Aerial Vehicle

RL Reinforcement Learning

DR Domain Randomization

MDP Markov Decision Process

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■ 1 Introduction

In the context of drone racing, where quadcopters are pushed to their physical and computational limits, there is a growing trend toward using neural networks for control. As demonstrated in [2], this approach has made it possible to outperform some of the world's most skilled human pilots.

These neural networks controllers are typically trained using RL algorithms. Due to the sample inefficiency of RL and the high cost of collecting real-world flight data, training is performed in simulated environments that provide a safe and scalable alternative. Despite this, the resulting control policies often perform worse than expected or fail to transfer to real-world scenarios altogether. This discrepancy in the policy's performance between simulation and reality is known as the sim-to-real gap, and it mainly arises from an undesired overfit of the learned policy to the simulated environment.

■ 1.1 Related works

This section should contain related state-of-the-art works and their relation to the author's work. We usually cite the original works like this [5]. You can also cite multiple papers at once like this [2], [4].

Demonstrating that Domain Randomization (DR) can significantly increase the adaptability and robustness of autonomous systems, the work of Ferede et al. [1] serves as the direct inspiration for this project. Their research showed that a DR-trained neural network controller could successfully generalize across various quadcopter platforms.

■ 1.2 Contributions

(Note: rather than contribute this work aims to reproduce the results from [1])

■ 1.3 Mathematical notation

It is a good practice to define basic mathematical notation in the introduction. See Table 1.1 for an example.

$\mathbf{x}, \boldsymbol{\alpha}$	vector, pseudo-vector, or tuple
$\hat{\mathbf{x}}, \hat{\boldsymbol{\omega}}$	unit vector or unit pseudo-vector
$\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3$	elements of the <i>standard basis</i>
$\mathbf{X}, \boldsymbol{\Omega}$	matrix
$\mathbf{I}$	identity matrix
$x = \mathbf{a}^\top \mathbf{b}$	inner product of $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$
$\mathbf{x} = \mathbf{a} \times \mathbf{b}$	cross product of $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$
$\mathbf{x} = \mathbf{a} \circ \mathbf{b}$	element-wise product of $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$
$\mathbf{x}_{(n)} = \mathbf{x}^\top \hat{\mathbf{e}}_n$	n^{th} vector element (row), $\mathbf{x}, \mathbf{e} \in \mathbb{R}^3$
$\mathbf{X}_{(a,b)}$	matrix element, (row, column)
x_d	x_d is <i>desired</i> , a reference
$\dot{x}, \ddot{x}, \dddot{x}, \ddot{\ddot{x}}$	1 st , 2 nd , 3 rd , and 4 th time derivative of x
$x_{[n]}$	x at the sample n
$\mathbf{A}, \mathbf{B}, \mathbf{x}$	LTI system matrix, input matrix and input vector
$SO(3)$	3D special orthogonal group of rotations
$SE(3)$	$SO(3) \times \mathbb{R}^3$, special Euclidean group

Table 1.1: Mathematical notation, nomenclature and notable symbols.

■ 2 Methodology

■ 2.1 Problem definition

We will define a domain as a subset of all the possible Markov Decision Process (MDP) environments. A standard domain is characterized by a 4-tuple $\mathcal{D} = (S, A, P, R)$ where:

- $\mathbf{S}$: denotes a set of states called the state space.
- $\mathbf{A}$: is the set of actions called the action space.
- $\mathbf{P}(\mathbf{s}'|\mathbf{s}, \mathbf{a})$: is the probability that a given action $a \in A$, at state $s \in S$ will lead to a new state $s' \in S$.
- $\mathbf{R}(\mathbf{s}', \mathbf{s}, \mathbf{a})$: is the immediate reward, or expected immediate reward, recived for transitioning from s to s' after performing action a .

From the source domain, $\mathcal{D}_s$, (i.e. simulation), we can control a set of N parameters of the variable aspects of each environment that define a configuration vector, $\xi = [\xi_1, \xi_2, \dots, \xi_N]$. Examples of ξ can include:

Physical properties	$\xi_{\text{mass}}, \xi_{\text{inertia}}, \xi_{\text{drag}}, \xi_{\text{gravity}}$
Perception properties	$\xi_{\text{sensor_noise}}, \xi_{\text{sensor_latency}}$
Viual properties	$\xi_{\text{lighting}}, \xi_{\text{texture}}, \xi_{\text{fov}}$
Scene properties	$\xi_{\text{gate_pos}}, \xi_{\text{gate_size}}, \xi_{\text{gate_num}}$

Each configuration is sampled from a randomization space, $\vec{\xi} \in \Xi \subset \mathbb{R}^N$, with a probability distribution $\vec{\xi} \sim p(\vec{\xi})$. Where the parameters are bounded to be within an interval $\xi_i \in [\xi_i^{\text{low}}, \xi_i^{\text{high}}]$ [3].

At the start of every training episode, a new environment insance is created by sampling a new configuration vector $\vec{\xi}$ from the distribution $p(\vec{\xi})$. Therefore, the agent will interact with a specific MDP environment, that is to say, a specific domain $\mathcal{D}_{\vec{\xi}}$. This way, the agent can collect data from a vareity of environments that otherwise would reamain unexplored.

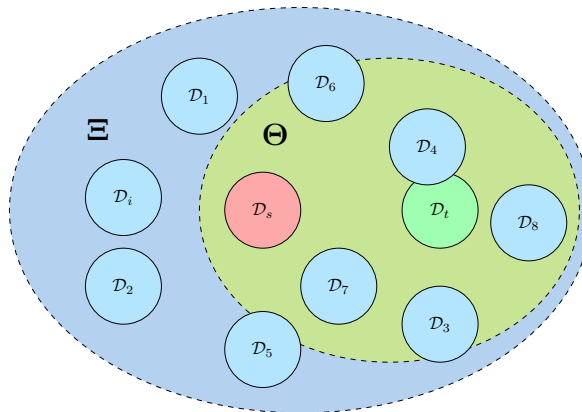


Figure 2.1: Red and green the source and target domains respectively. In blue, differents instances of the randomized source space (all represented domains with the same size, N).

Hereby, during training, the neural network weights and biases θ are optimized to maximize the expected reward R across a distribution of randomized domains. With th objective

of finding the optimal, θ^* .

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{\tilde{\xi} \sim \Xi} [\mathbb{E}_{\pi_{\theta}, \tau \sim D_{\xi}} [R(\tau)]] \quad (2.1)$$

And it will be able to react to certain unseen variations of the randomization space. More specifically, from the

■ 2.2 Quadcopter model

■ 2.3 RL set-up

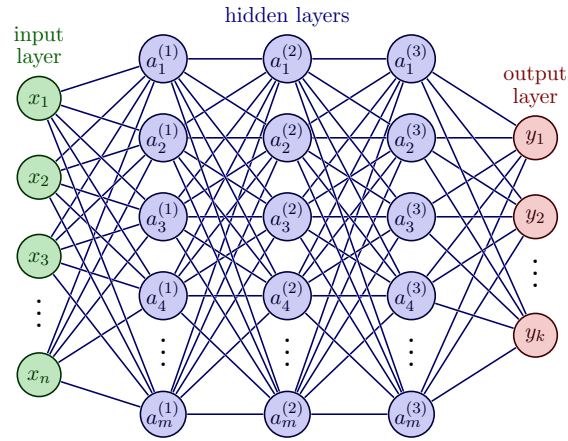


Figure 2.2: Red and green the source and target domains respectively. In blue, different instances of the randomized source space (all represented domains with the same size, N).

■ 2.4 DR configurations

3 Results

Summarize the achieved results. Can be similar as an abstract or an introduction, however, it should be written in past tense.

4 Conclusion

Summarize the achieved results. Can be similar as an abstract or an introduction, however, it should be written in past tense.

■ 5 References

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A Appendix A